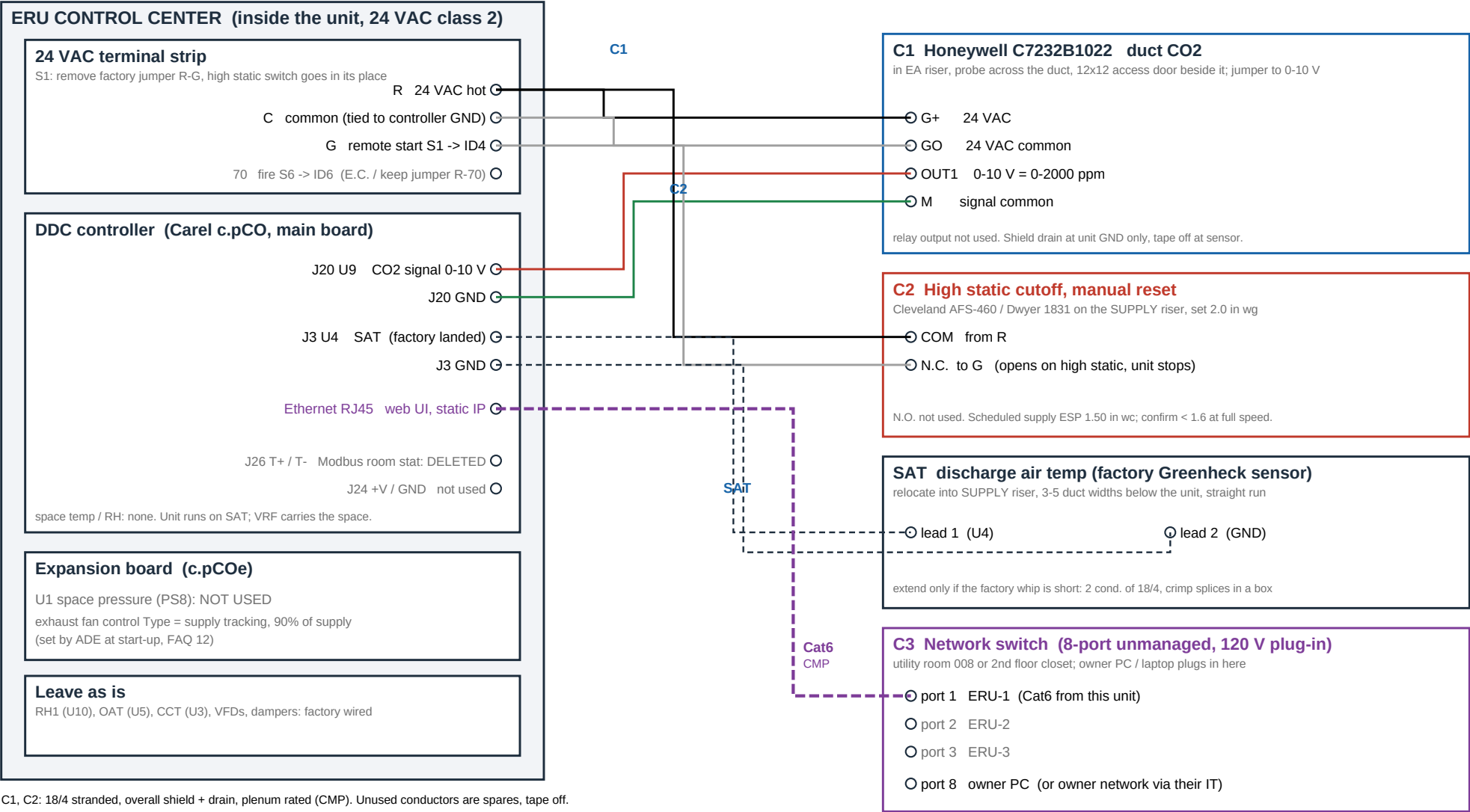


Northern Wolves AC | IPA Church, 310A S. Oyster Bay Rd, Syosset NY | ref. Greenheck wiring diagrams G31 (controller) and Y07 (expansion), sent by ADE 09/30/2026



C1, C2: 18/4 stranded, overall shield + drain, plenum rated (CMP). Unused conductors are spares, tape off.

Shield drain on controller GND at the unit only. Never in a conduit with 208 V. Runs 20-60 ft.

C3: Cat6 CMP, one home run per unit to the switch, no splices, under 328 ft; RJ45 at both ends.

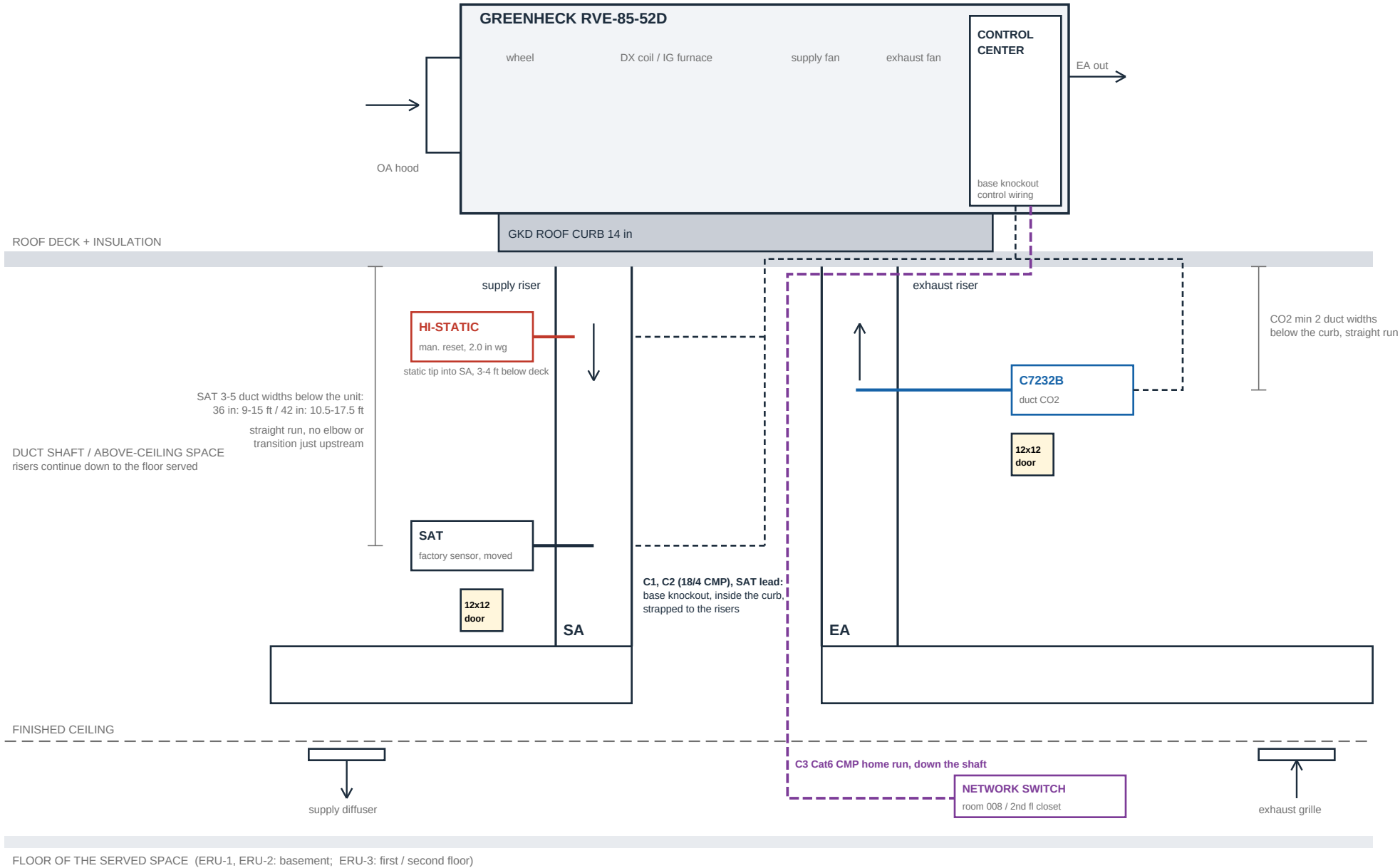
Route: unit base control knockout -> inside the curb -> down the duct shaft. No roof / wall penetrations.

Fire S6 (R-70) and duct smoke detectors: E.C. scope, already installed; do not disturb.

Cable	Device	BLK	WHT	RED	GRN	Length
C1	duct CO2 Honeywell C7232B1022	R -> G+	C -> GO	OUT1 -> U9	M -> GND (U9)	20-60 ft
C2	high static cutoff, manual reset	R -> COM	N.C. -> G	spare	spare	< 20 ft
SAT	discharge air temp (factory)	lead 1 -> U4	lead 2 -> GND	-	-	factory lead
C3	Cat6 to network switch	controller Ethernet port -> switch port 1/2/3				to 008

G31: C is tied to controller GND, so M on the GND beside U9 = M on C as the C7232 legend shows.

DELETED: HU-226 temp/RH, PR-274, A-306 probe, tubing, A-489 pickup, 3 remote displays.



PLACEMENT NOTES

1. CO2 on the exhaust riser, first straight section reachable from inside: ERU-3 at the 2nd floor ceiling (grid 8.7-9.2), ERU-1 / ERU-2 at the top of the shaft under the deck. Probe across the centerline, gasketed hole, 12x12 door within 2 ft.
2. SAT (Greenheck FAQ 6): in the supply duct 3-5 duct widths downstream of the unit, away from elbows, transitions and major turns. Probe across the centerline, access door beside it.
3. High static cutoff on the supply riser 3-4 ft below the deck, static tip only (low port open to the shaft). Manual reset, in series with S1 R-G (factory jumper removed). Nothing inside the curb or on the roof section of the ducts.
4. Web UI: one Cat6 home run per unit to the switch; ADE sets a static IP per unit on 10/29. Space temp/RH, building pressure sensor, outdoor probe and remote displays are deleted (client approved 09/30).

#	Item	Qty	Where	Approx. each	Approx. total
1	Honeywell C7232B1022 duct CO2 sensor with LCD (Greenheck 472066)	3	SupplyHouse, Kele, parts-hvac	\$280-570	\$850-1,710
2	Manual reset air pressure switch, SPDT (Cleveland AFS-460 or AFS-305; Dwyer 1831)	3	SupplyHouse, Cleveland dist.	\$120-170	\$360-510
3	Static pressure tip for the switch (Dwyer A-301 or equal) + 10 ft 1/4 in tubing each	3	Kele, SupplyHouse	\$20-30	\$60-90
4	18/4 stranded shielded plenum (CMP) cable, 500 ft spool (C1 CO2, C2 switch, SAT ext.)	1	CablingPlus, CableWholesale	\$120-180	\$120-180
5	Cat6 plenum (CMP) solid cable, 1000 ft box (C3, one home run per unit)	1	electrical / datacom supply	\$200-300	\$200-300
6	Cat6 RJ45 plugs (bag of 50) + 3 ft patch cords x4	1 lot	electrical / datacom supply	\$40-60	\$40-60
7	8-port unmanaged gigabit switch, metal case (Netgear GS308 or equal)	1	any	\$30-60	\$30-60
8	12x12 insulated duct access doors (CO2 on EA riser, SAT on SA riser)	6	our shop / Ventlok	\$35-60	\$210-360
9	Misc: J-hooks / bridle rings, plenum cable ties, knockout bushings, 4x4 box + gel crimp	1 lot	stock		\$100-150
	splices for SAT lead extension (only if factory whip is short), labels, duct sealant				
	TOTAL MATERIAL, 3 units				\$1,970-3,420

NOT PURCHASED (deleted with the client's approval):

- Mamac HU-226 duct temp/RH - no duct RH input on this program; unit controls on SAT, VRF carries the space.
- Mamac PR-274 transducer, Dwyer A-306 outdoor probe, A-489 pickup, pressure tubing - exhaust fan set to track supply at 90%.
- 3 Greenheck remote displays (~\$1,400 each) - replaced by the unit web UI over Ethernet (no license).
- ADE reduced sensor package (\$10,500 + tax) - we buy the parts; programming is in the PO start-up visit on 10/29.

BY OTHERS / CHECK:

- Owner PC or laptop for the web UI, and any connection to the owner's network / remote access: owner and their IT.
- 120 V outlet at the network switch location (utility room 008 or 2nd floor closet): confirm one exists, else E.C.
- Smoke detectors and fire input S6: E.C., already installed. Existing SAT sensors: in the units, reused.
- Prices are Sept 2026 web prices before tax and shipping; links in the engineering note "Where to buy".